In22-S5-CS3501 Data Science and Engineering Project
Department of Computer Science and Engineering
University of Moratuwa

Development of a Natural-Sounding Sinhala Text-to-Speech Synthesis System

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Group ID: ***18***
Project ID: ***P15***
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1. Executive Summary:

We will build a working neural Text-to-Speech (TTS) application that converts written Sinhala text into natural sounding speech. Although Sinhala is spoken by about 17 million people, it has been left out of major industry TTS efforts. We will train and fine-tune modern neural TTS architectures (VITS and F5-TTS) on different openly licensed Sinhala TTS datasets. The system will be delivered as a web application and evaluated with both native-speaker Mean Opinion Score (MOS) studies and automatic intelligibility metrics (Whisper-based WER), compared against the community baselines we verified to exist.

2. Problem Statement:

Speech interfaces and accessibility tools depend on high-quality TTS, but a massive technology gap exists for Sinhala. Our verified preliminary findings show: 1. **No open industry model exists.** The sole offering is Microsoft Azure's paid cloud API which cannot run offline. 2. **Available data is scattered.** Datasets are often poorly transcribed or lack reliable metadata. 3. **Existing community models are unevaluated.** There is no published quality evaluation of existing Sinhala TTS checkpoints. 4. **Code-switching is unaddressed.** Everyday written Sinhala freely mixes English words, yet no system handles English islands inside Sinhala text.

3. Data Description:

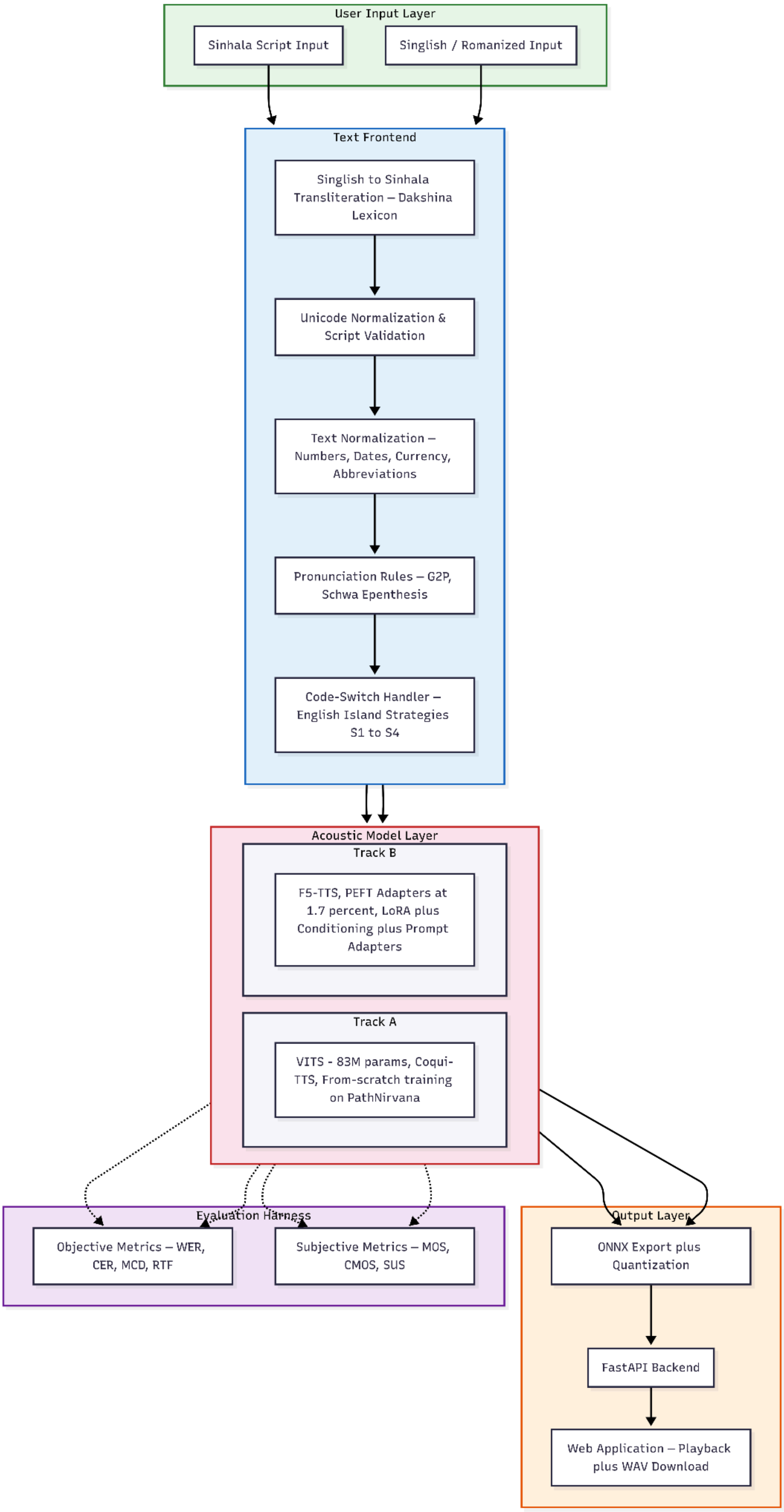
Corpus	Source	Measured Contents	Type / Format	License	Role
PathNirvana (pnfo)	pnfo repo	6,386 clips, 13.61 h, 22.05 kHz studio quality; male 11.58 h, female 2.03 h	Audio (wav) + LJSpeech-style metadata	GPL-3.0 (+ non-offensive-use clause)	Primary training corpus
OpenSLR SLR30	openslr 30	2,064 clips, 12 speakers, 3.38 h, 48 kHz, only 1,251 clips (~2 h) transcribed	Audio (wav) + transcript lines	CC BY-SA 4.0	Multi-speaker experiments
Safnas Kaldeen	Kaggle	4 native speakers, phonetically balanced sentences (UoM FYP 2025)	Audio + transcripts	Apache-2.0	Speaker-variety fine-tuning

4. Methods:

4.1 System Overview & Diagram

The system comprises a text frontend for normalization, an acoustic layer (VITS and F5-TTS models), and a FastAPI web backend. A shared evaluation harness objectively scores all variants.

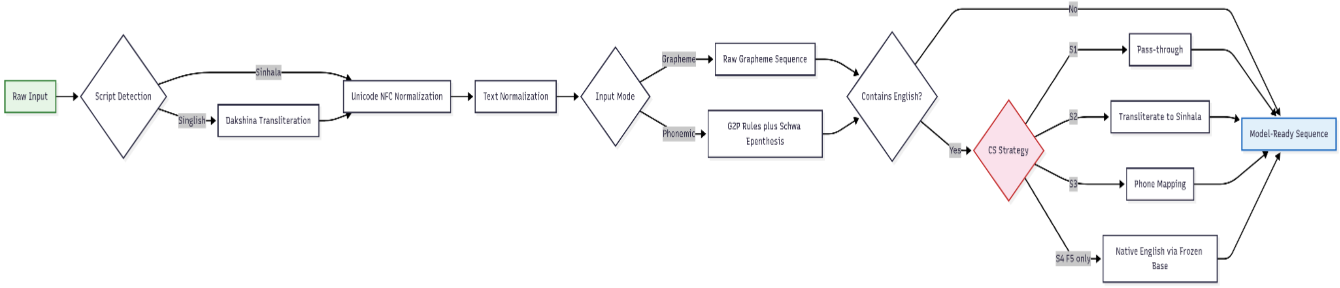
System Architecture



4.2 Data Preparation & Text Frontend

Audio undergoes uniform resampling (22.05kHz/24kHz) and EBU R128 loudness normalization. Transcripts undergo Unicode NFC normalization and are validated against a Sinhala Whisper ASR model. The frontend performs rule-based expansion of non-standard words (numbers, dates) using Google's grammar rules, supports varied input representations (graphemes vs phonemes), and enables Singlish transliteration via the Dakshina lexicon.

Text Frontend Pipeline



4.3 Model Development & Code-Switching

Track A, VITS: Trained on PathNirvana male subset (11.58 h) up to 100k+ steps, followed by female voice fine-tuning. **Track B, F5-TTS:** Parameter-efficient fine-tuning (LoRA) freezing the multilingual base to preserve English competence. For **Code-Switching**, we curate a test set from NSINA and compare strategies: (S1) naive Latin pass-through, (S2) Sinhala grapheme transliteration, (S3) hybrid phone mapping, and (S4) native English islands on F5-TTS.

5. Evaluation Plan

Test Sets: FLEURS si_lk (general intelligibility), Normalization Stress Set, Rare-Syllable Set, and Code-Switched Set.

Planned Experiments:

ID	Experiment	Description
E1	Impact of text normalization	Synthesize normalization stress set with normalizer enabled vs. disabled; measure WER and listener preference
E2	Impact of pronunciation rules	Grapheme input vs. phonemic input derived from letter-to-sound rules; compare WER on rare-syllable set
E3	Code-switching strategies	S1 vs S2 vs S3 (vs S4 on Track B) on code-switched set, with island-level vs carrier-level WER + acceptability ratings
E4	Data scale	Voices trained on 2 h vs 6 h vs 13.6 h to characterize quality to data curve

6. Expected Outcomes and Success Criteria:

ID	Deliverable	Description
D1	Working Sinhala TTS web application	Primary outcome, male and female voices, normalization frontend, and Singlish input
D2	Public evaluation suite	Four test sets, scoring scripts, and the full benchmark table over all systems
D3	Model checkpoints and training recipes	For both VITS and F5-TTS tracks, with text-normalization/G2P frontend as standalone open-source module
D4	Data manifests and preprocessing pipeline	Reproducing our cleaned corpora from the public originals
D5	Research paper draft	Built on benchmark (C3) and code-switching (C4), candidate venues: ICTer, MERCon, SLTU/ComputEL
D6	Module documentation	Final report, demonstration, and maintained logbook/Jira records

7. Division of work (individual responsibilities)

- **Nethmina L.T.H [Data & Frontend]:** Corpus pipeline, normalization grammar, Singlish, CS test-set.
- **Pahanga G.H.L [Modelling]:** VITS training, F5-TTS adapter track, Colab/Kaggle management, model export.(ONNX)
- **Pabasara W.G.K. [Evaluation & App]:** Benchmark harness, MOS study execution, Web app deployment.

8. Preliminary Bibliography:

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6. Chen et al., "F5-TTS," arXiv 2024.
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8. Joshi et al., "Code-Mixed TTS Synthesis," arXiv 2023.
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